Edwin Binu

AI Engineer | AI Governance, LLM Applications, Workflow Automation & Agentic Systems

Dubai, UAE (UAE Golden Visa – no sponsorship required) | +971 50 258 0178 | edwinbinu07@gmail.com | linkedin.com/in/edwin-binu-d11041 | github.com/eb4005 | eb4005.github.io

Professional Summary

AI Engineer specializing in AI governance, LLM-powered applications and workflow automation for regulated environments. At ioHealth, leads development of Ethos, an enterprise AI governance platform that connects AI agent registration, runtime policy decisions (allow / block / escalate / hold), Responsible AI evaluation, human-in-the-loop review and audit evidence for healthcare AI agents. Hands-on across the stack: Java 21/Spring Boot and React for platform engineering; Python, LangChain, FastAPI and RAG for LLM applications; n8n, webhooks and GitHub automation for production workflows. Built PHI/PII detection and guardrail benchmarks (F1 0.85, Presidio + GLiNER). MSc in Artificial Intelligence, Heriot-Watt University. Strong communicator across technical and non-technical audiences.

Technical Skills

- **AI Governance & Safety:** AI Governance, Responsible AI (RAI) Evaluation, Human-in-the-Loop (HITL) Oversight, AI Guardrails, Policy/Rule Engines, Audit Trails, PHI/PII Detection, Named Entity Recognition (NER), Prompt Injection Detection, Microsoft Presidio, GLiNER, Llama Prompt Guard 2, ProtectAI
- **LLM & Agentic AI:** Large Language Models (LLM), RAG Pipelines, Multi-Agent Orchestration, LangChain, LangGraph, CrewAI, Prompt Engineering, Context Engineering, Anthropic Claude API, OpenAI API, Groq (Llama 3, Mixtral, Gemma), Hugging Face, Vertex AI, Multi-Provider Abstraction Layers, FAISS, pgvector, SentenceTransformers, Tavily
- **Workflow Automation:** n8n (workflow design, webhooks, production automations), Webhook Orchestration, GitHub API, GitHub Apps, Chrome Extensions (Manifest V3), Workflow Version Control, Automated Scorecards & Reporting
- **AI Observability & Evaluation:** Evaluation Harnesses, LLM-as-a-Judge, Benchmarking (Precision/Recall/F1), Prompt Versioning, Inference Latency Monitoring, Output Quality Scoring
- **Backend & Frontend:** Java 21, Spring Boot, Spring Security, JPA/Hibernate, Python, FastAPI, Node.js, Express.js, REST APIs, GraphQL, React, TypeScript, JavaScript, Vite, Tailwind CSS, TanStack Query, shadcn/ui
- **Security & Identity:** Keycloak, JWT, RBAC, Multi-Tenant Access Control, API-Key Authentication, Rate Limiting, Audit Logging, Least-Privilege, Secure Coding
- **Data:** MySQL, PostgreSQL, Supabase, MongoDB, Redis, SQL, Flyway, Relational Data Modelling, Python ETL, pandas, NumPy
- **DevOps & Testing:** Docker, Docker Compose, GitHub Actions, CI/CD, Git, AWS, Azure, JUnit, Testcontainers, Playwright, Postman, Render

Professional Experience

ioHealth – AI Automation and Governance | ***Jun 2026 – Present*** | ***Dubai, UAE***

Project Lead – Ethos AI Governance Platform (React, Java 21, Spring Boot, MySQL, Keycloak, Flyway, n8n)

- Lead end-to-end development of Ethos, an enterprise AI governance platform for registering, monitoring and enforcing policy compliance of AI agents; evolved it from a frontend prototype into a full-stack application and led its rebrand from Comply to Ethos.
- Platform spans an agent registry with n8n workflow bindings and GraphQL agent discovery; a runtime decision API returning allow, block, escalate or hold for agent actions; asynchronous Responsible AI evaluations tracked by correlation IDs; and Pivots, a human-in-the-loop review queue retaining evidence snapshots, reviewer attribution and decision history.
- Building multi-tenant, organization-scoped Pivots: per-organization review queues and history, object-level authorization and a platform super-admin view, with controlled tenant onboarding and cleanup of legacy organization records.
- Drove a production-readiness sprint (UI consolidation, security controls, KPI frameworks, validation architecture); led a platform audit and remediated compliance-scoring and rule-evaluation defects.
- Designed an AI governance framework of 7 agent-type risk profiles (e.g., PHI Handler, LLM Content Generator, Clinical Decision Support), each with defined security checks and KPIs.
- Delivered rule-monitoring and incident-enforcement features and presented the release demo to stakeholders; leading escalation policies, automated incident reporting, PHI assessment for agent runs and migration of workflows from n8n to an in-house workflow platform.

AI Guardrails and Runtime Governance Engine (n8n, Supabase, PostgreSQL, Python, LLM)

- Built core n8n pipelines for payload extraction, policy ingestion (converting PDF policy documents into structured, document-grounded rules) and a validation gate returning block, escalate or allow decisions for AI agent actions.
- Implemented an evaluation-results database and automated scorecard workflows; benchmarked rule coverage against a 111-rule ground-truth dataset.

Responsible AI and Guardrail Tool Evaluation (Python, Presidio, GLiNER, Llama Prompt Guard 2, Hugging Face, Docker)

- Led evaluation of open-source Responsible AI and guardrail tools for a client deliverable, producing precision, recall and F1 benchmarks on public Hugging Face datasets.
- Achieved an F1 score of 0.85 for PHI/PII detection using Presidio with GLiNER NER; identified multilingual prompt-injection detection gaps (Arabic, German).
- Built a PHI redaction demo application and n8n webhook workflow, with Docker deployment artifacts for internal hosting.

Workflow Version Control and Compliance Automation (Node.js, Express, PostgreSQL, Docker, Chrome Extension, GitHub API)

- Built and deployed to production a Chrome extension and Node.js backend that automatically version-control n8n workflows to GitHub as branches and pull requests, with full audit logging in PostgreSQL.
- Added LLM-generated pull-request descriptions with checksum-based caching; owned the version-control metric used in AI agent compliance scoring.
- Applied security hardening (timing-safe authentication, CORS controls, rate limiting, credential rotation) and wrote deployment and user documentation for approximately 70 team members.

Heriot-Watt University – Teaching Assistant | ***Jan 2026 – May 2026*** | ***Dubai, UAE***

Advanced Network Security and AI Labs

- Led workshop-style lab sessions for postgraduate and undergraduate cohorts; supervised Docker and Ubuntu VM lab environments, translating complex AI and security architectures into hands-on guidance.
- Taught secure system configuration, least-privilege access and JWT-based authentication relevant to production AI system hardening.

Zenardy – Associate Consultant | ***Feb 2023 – Aug 2023*** | ***Chennai, India***

Enterprise AI Workflow Design

- Embedded with enterprise clients to translate business workflows into automated backend solutions; led design workshops and POC sessions, communicating trade-offs to non-technical stakeholders at each milestone.
- Built Python ETL pipelines and automated workflow orchestrations, improving data consistency and reducing manual effort across enterprise systems.

Projects

MediAssist AI – Medical ChatBot with RAG and LLM | ***2025***

github.com/eb4005/AI-Medical-ChatBot

- Built a medical AI assistant using LangChain, Groq LLMs (Llama 3, Mixtral, Gemma) and FAISS, with a full RAG pipeline interpreting uploaded patient PDFs (blood tests, pathology reports, visit summaries).
- Designed three conversation modes (Standard Chat, Diagnostic Analysis, Jargon Summarisation) with per-mode prompts, Hugging Face BGE embeddings and a Tavily web-search fallback for grounded answers.

AI Resume Scanner – Full-Stack LLM Hiring Tool | ***2025***

github.com/eb4005/AI-Resume-Scanner-

- Built and deployed on Render a resume-screening platform (React 19, TypeScript, Express.js, Groq LLM) with batch analysis of up to 10 resumes, match scores (0–100%), hire/reject recommendations and skills-gap analysis.
- Designed a REST API with Multer uploads and pdf-parse extraction returning a structured JSON schema of scores, skills and recommendations.

Enterprise AI Multi-Agent Orchestration System | ***Mar 2025 – May 2025***

github.com/eb4005

- Built a multi-agent system with Python and FastAPI services implementing policy-based agent routing, tool invocation pipelines and agent lifecycle observability, with a multi-provider abstraction layer for Hugging Face models.
- Implemented RAG with SentenceTransformers and FAISS (pgvector-compatible); deployed via Docker Compose with GitHub Actions CI/CD.

Education

Heriot-Watt University – MSc in Artificial Intelligence | ***Sept 2025*** | ***Dubai, UAE***

Anna University – BE in Computer Science, GPA: 8.5 / 10.0 | ***July 2023*** | ***Chennai, India***

Certifications

IBM Certified: Machine Learning with Python | In progress: Anthropic Claude AI Certification, DeepLearning.AI Agentic AI Specialisation

Languages

English (Fluent) | Malayalam | Tamil | Hindi | Arabic

Publications

"Trash Image Detection Using Autoencoder" – International Conference on Computing, Communication and Security (ICCCS), 2023